KILNDECK · FEASIBILITY ASSESSMENT

KilnDeck feasibility assessment

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/kilndeck		
Source	docs/08-feasibility.md @ 9f92bd8		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-30	Amish Chadha	Manual handling and wind limit follow the decisions of 2026-10-30; cost USD 2,905
0.1	2026-10-28	Amish Chadha	First issue (kit 1.8 upgrade)

KilnDeck feasibility assessment

On paper the rolling bridge deck can be designed, made and fitted with local skills and stock steel for about USD 2,905; what is not yet known is whether real kiln walls carry it, whether firemen will use it, and whether it stays cool enough by day. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	Strength, stiffness, travel force, parking and storm hold all met on paper with wide margins; floor heat not met in full sun	High	PR-007 not met in full sun; PR-012 (33 kg) not met by the end truss modules, 34.4 and 34.8 kg	KND-STM-001, KND-CAL-001
Materials	S275 hollow section, channel, plate and sheet; calcium silicate board; aluminium sheet; silicone sleeves	Medium	Board compressive strength and heat-rated paint to confirm when bought	KND-DEC-001 items 4, 11
Manufacturing and assembly	Cut, drill and stick weld in a small workshop; bolted together on the kiln over the cold zone; no piece over 33 kg except the end truss modules	High	The end truss modules are 1.4 to 1.8 kg over the 33 kg limit (D14, D16); work at the kiln edge during assembly	KND-BLD-001, `cad/fidelity.json`
Power and controls	Manual only: hand wheel, lock pin, crank; no power on the kiln	High	Wind above 7 m/s is outside the working limit; clamps and chains are set (D11, decided 2026-10-30)	KND-CAL-001 s10, s15
Supply of parts	Gate wheels, UC bearings, 08B chain and sprockets, hand wheel, rigging are common catalogue items	Medium	Wheel rating (650 kg, D15) and groove widths to confirm; rigging prices are estimates	BOM lines 5, 10, 18
Site	Needs straight wall tops at least 0.6 m wide, level with the crust within 100 mm, and room beyond the walls for tie-down anchors	Low	Not yet surveyed; the curved kiln ends are not served	KND-PRB-001; STM S9, S21
Adoption	Firemen must accept working from a deck over the fire, and owners the cost and track handling	Low	Not yet asked; first co-design task	KND-PRB-001 co-design checklist

Ratings: High, Medium, Low.